Inferring Hidden Competitive Intelligence in Pharmaceutical Negotiations: Quantum-Trained Boltzmann Machines for Strategic Advantage in PBM Rebate Contracts

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Abstract

In pharmaceutical benefit manager (PBM) rebate negotiations, the greatest strategic disadvantage is information asymmetry—not knowing competitor rebate percentages, PBM margins across manufacturers, or portfolio-level deal structures. This paper presents a groundbreaking application of quantum-trained Restricted Boltzmann Machines (RBMs) that transforms these unknown variables from liabilities into strategic advantages. The hidden nodes in RBMs naturally learn to represent unobservable market information, effectively inferring competitor rebates and PBM strategies from observable market patterns. When trained using D-Wave's quantum annealing hardware through their new PyTorch-integrated toolkit, these models can discover hidden correlations that classical methods miss—such as detecting portfolio deals from seemingly unrelated formulary changes or inferring rebate ranges from market share movements. Unlike traditional game-theoretic approaches that require complete information, our framework thrives on information asymmetry, learning latent representations that capture the effects of unknown variables on market dynamics. The quantum training process enables the discovery of rare but highly valuable strategic patterns hiding in narrow energy wells, patterns that classical gradient descent would never find. For pharmaceutical executives, this represents a paradigm shift: the very information your competitors guard most closely can be statistically

inferred from market observations, providing quantifiable competitive intelligence for negotiation strategy. We discuss how this approach moves beyond traditional anomaly detection to strategic navigation, where the energy landscape represents negotiation possibilities and quantum tunneling reveals paths to superior agreements that classical negotiation strategies would never discover.

1 Introduction

The pharmaceutical supply chain in the United States operates through a complex web of negotiations between drug manufacturers, pharmacy benefit managers (PBMs), and health plans. The rebate system, where manufacturers provide retrospective discounts to PBMs in exchange for favorable formulary placement, creates an opaque market with significant information asymmetries. These negotiations represent a multi-agent optimization problem where each party seeks to maximize their objectives while maintaining system stability.

The critical challenge in PBM negotiations is not what you know—it's what you don't know. Competitor rebate percentages, PBM margins across manufacturers, portfolio deal structures, and strategic intentions remain hidden, yet these unknown variables often determine negotiation outcomes. Traditional analytical approaches fail precisely because they cannot handle these information gaps.

Boltzmann machines, particularly Restricted Boltzmann Machines (RBMs), offer a revolutionary solution: their hidden nodes naturally learn to represent these unobservable variables. When implemented on quantum annealing hardware through D-Wave's new PyTorch-integrated toolkit, these models could gain the ability to discover hidden market patterns through quantum mechanical effects, revealing competitive intelligence invisible to classical analysis. This transforms information asymmetry from a strategic weakness into a quantifiable, actionable advantage.

2 Executive Summary: The Competitive Intelligence Advantage

2.1 The Information Asymmetry Problem

In PBM rebate negotiations, critical information remains hidden:

- **Competitor Rebates**: Your competitor's exact rebate percentage

with a PBM is their most guarded secret, yet it determines your formulary position

- **Portfolio Deals**: Cross-product agreements that link seemingly unrelated therapeutic areas
- **PBM Margins**: The true profitability PBMs achieve across different manufacturers
- **Strategic Intent**: Whether competitors are buying share, defending position, or preparing for new launches

Current approaches—hiring consultants, purchasing syndicated data, or relying on relationships—provide incomplete, often outdated intelligence. Meanwhile, every basis point of rebate disadvantage costs millions in lost revenue.

2.2 The Hidden Node Solution

RBMs contain two layers: visible nodes (what you can observe) and hidden nodes (latent variables the model learns). Critically, these hidden nodes learn to represent information you cannot directly observe:

Observable (Visible Nodes)	**Inferred (Hidden Nodes)**
Formulary positions	Competitor rebate ranges
Market share changes	Portfolio deal presence
List price movements	PBM margin satisfaction
Earnings call language	Strategic intentions
Biosimilar launches	Defensive rebating levels

The hidden nodes do not magically reveal exact competitor rebates. Instead, they learn statistical patterns: when observable market behaviors X, Y, and Z occur together, there is high probability of a 45-50% competitor rebate. This probabilistic intelligence is more valuable than point estimates because it quantifies uncertainty and indicates confidence levels.

2.3 The Quantum Training Advantage

Classical training methods get trapped in obvious patterns. Quantum annealing, through D-Wave's platform, enables:

- **Discovery of Rare Patterns**: Quantum tunneling finds correlations in narrow probability wells—such as detecting early signals of competitive strategy shifts

- **Multi-Modal Learning**: Simultaneously learning different types of deals (straight rebates, portfolio agreements, value-based contracts) that exist in separate energy minima
- **Escape from Training Bias**: Classical methods reinforce historical patterns; quantum training can discover novel strategic configurations

2.4 Competitive Moat Through Data

Even if competitors adopt similar quantum RBM approaches, your competitive advantage persists because:

- Your hidden nodes train on your proprietary negotiation history
- Your win/loss patterns teach the model your specific competitive position
- Your model learns to detect patterns most relevant to your portfolio and strategy

2.5 ROI and Strategic Value

For a large pharmaceutical manufacturer:

- **Improved Negotiation Outcomes**: Even 1-2% better rebate efficiency across a $10B portfolio represents $100-200M annual value
- **Faster Market Response**: Detecting competitive moves weeks earlier enables proactive rather than reactive strategies
- **Portfolio Optimization**: Understanding cross-product leverage improves entire portfolio positioning
- **Risk Mitigation**: Early warning signals for formulary exclusions or competitive disruptions

2.6 Implementation Path

With D-Wave's Ocean toolkit and PyTorch integration, implementation requires:

1. Historical data collection (formulary changes, market shares, pricing)
2. Model training using quantum annealing for hidden node optimization

3. Validation against known historical outcomes

4. Forward-looking inference for current negotiations

The question is not whether to pursue this capability, but whether your competitors will develop it first.

3 Theoretical Background

3.1 Classical Boltzmann Machines

A Boltzmann machine defines an energy function over a set of binary units:

$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{v}^T\mathbf{W}\mathbf{h} - \mathbf{v}^T\mathbf{L}\mathbf{v} - \mathbf{h}^T\mathbf{J}\mathbf{h} - \mathbf{a}^T\mathbf{v} - \mathbf{b}^T\mathbf{h} \tag{1}$$

where $\mathbf{v}$ represents visible units, $\mathbf{h}$ represents hidden units, $\mathbf{W}$ captures visible-hidden interactions, $\mathbf{L}$ represents visible-visible connections, $\mathbf{J}$ represents hidden-hidden connections, and $\mathbf{a}$, $\mathbf{b}$ are bias terms.

The probability of any configuration follows the Boltzmann distribution:

$$P(\mathbf{v}, \mathbf{h}) = \frac{1}{Z}\exp(-E(\mathbf{v}, \mathbf{h})/T) \tag{2}$$

where Z is the partition function and T is the temperature parameter.

3.2 Restricted Boltzmann Machines

Restricted Boltzmann Machines (RBMs) eliminate intra-layer connections ($\mathbf{L} = 0$, $\mathbf{J} = 0$), creating conditional independence:

$$P(\mathbf{h}|\mathbf{v}) = \prod_i P(h_i|\mathbf{v}) \tag{3}$$

$$P(\mathbf{v}|\mathbf{h}) = \prod_j P(v_j|\mathbf{h}) \tag{4}$$

This restriction enables efficient parallel sampling and training through contrastive divergence, at the cost of not directly modeling correlations within layers.

3.3 The Critical Insight: Hidden Nodes as Competitive Intelligence

The transformative insight for PBM negotiations is that hidden nodes in RBMs naturally learn to represent unobservable market information. In traditional machine learning, hidden layers are often treated as abstract feature extractors. In the context of PBM negotiations, these hidden nodes become concrete representations of the information asymmetries that define market power:

- **Hidden Node Activation → Competitor Rebate Range**: Specific patterns of hidden node activation correspond to probabilistic estimates of competitor rebate levels
- **Cross-Node Correlations → Portfolio Deals**: Correlations between hidden nodes reveal linkages across therapeutic areas
- **Temporal Changes → Strategic Shifts**: Evolution of hidden node patterns signals competitor strategy changes before they become observable

This is not merely pattern recognition—it is the systematic inference of strategically critical information from incomplete observations. The hidden nodes learn the statistical signatures of unobservable variables, transforming market opacity into probabilistic intelligence.

3.4 Quantum Annealing

Quantum annealing implements optimization through a time-dependent Hamiltonian:

$$H(t) = A(t)H_{\text{transverse}} + B(t)H_{\text{problem}} \tag{5}$$

where $H_{\text{transverse}}$ introduces quantum fluctuations enabling tunneling, H_{problem} encodes the optimization objective, and the annealing schedule has $A(t)$ decreasing while $B(t)$ increases over time $t \in [0, \tau]$.

The quantum advantage for competitive intelligence is profound: quantum tunneling enables discovery of hidden correlations that exist in narrow probability wells—exactly where the most valuable competitive insights hide.

4 PBM Negotiations as Energy Landscapes

4.1 The Paradigm Shift

Traditional applications of Boltzmann machines in healthcare focus on anomaly detection, where:

- Low energy states represent normal, healthy operation
- High energy states indicate anomalies requiring attention
- The goal is to detect when the system deviates from learned equilibria

For PBM negotiations, we propose an inverted paradigm:

- Current negotiated agreements represent local energy minima
- Better potential agreements exist as unexplored minima
- The goal is to navigate from current equilibrium to superior equilibria

4.2 Multi-Agent Energy Functions

Each stakeholder in the negotiation maintains their own energy function. For a manufacturer:

$$E_{\text{mfr}}(\mathbf{s}) = -\pi(\mathbf{s}) - \alpha \cdot \text{access}(\mathbf{s}) - \beta \cdot \text{strategic}(\mathbf{s}) + \gamma \cdot \text{risk}(\mathbf{s}) \tag{6}$$

where $\mathbf{s}$ represents the negotiation state, π is profit, and α, β, γ are weighting parameters.

For the PBM:

$$E_{\text{pbm}}(\mathbf{s}) = -\text{rebate}(\mathbf{s}) - \text{fees}(\mathbf{s}) - \delta \cdot \text{plan_satisfaction}(\mathbf{s}) + \epsilon \cdot \text{complexity}(\mathbf{s}) \tag{7}$$

The joint energy landscape for negotiation becomes:

$$E_{\text{joint}}(\mathbf{s}) = E_{\text{mfr}}(\mathbf{s}) + \lambda E_{\text{pbm}}(\mathbf{s}) + \mu E_{\text{system}}(\mathbf{s}) \tag{8}$$

where λ and μ balance stakeholder interests and system-wide efficiency.

5 Observable and Hidden Variables in PBM Markets

5.1 The Competitive Intelligence Framework

The power of RBMs in PBM negotiations lies in their ability to infer what competitors and PBMs will never tell you. Every negotiation is influenced by information you cannot access—competitor rebates, PBM margins, strategic intentions—yet these hidden variables leave fingerprints in observable market behaviors. RBMs learn to read these fingerprints.

5.2 Visible Layer: Market Observables

The visible units in our Boltzmann machine correspond to data you can obtain through public sources, syndicated data, or market observation:

$$\mathbf{v} = \begin{bmatrix} \text{WAC prices (public)} \\ \text{Formulary positions (accessible)} \\ \text{Market shares (IQVIA data)} \\ \text{Patient copays (benefit designs)} \\ \text{Competitor presence (public)} \\ \text{Earnings guidance (public)} \\ \text{Prescriber patterns (claims data)} \end{bmatrix} \tag{9}$$

5.3 Hidden Layer: Competitive Intelligence

The hidden units learn to represent the information that determines negotiation outcomes but remains invisible:

$$\mathbf{h} = \begin{bmatrix} \text{Competitor rebate percentages} \\ \text{PBM margin requirements} \\ \text{Portfolio deal structures} \\ \text{Future competitive entries} \\ \text{Strategic reserve prices} \\ \text{Biosimilar defense thresholds} \\ \text{Value-based contract presence} \end{bmatrix} \tag{10}$$

5.4 Learning Competitive Patterns: Concrete Examples

5.4.1 Example 1: Inferring Competitor Rebates

When Competitor A achieves preferred formulary status despite a 5% higher list price than your product, the hidden nodes learn this pattern:

$$P(\text{Rebate}_A > \text{Rebate}_{yours} + 10\% | \text{Observable patterns}) > 0.85 \tag{11}$$

The model learns that specific combinations of observables (formulary advantage despite price disadvantage + stable market share + PBM earnings satisfaction) indicate high probability of significant rebate differential.

5.4.2 Example 2: Detecting Portfolio Deals

Hidden nodes identify portfolio linkages by learning correlations across therapeutic areas:

$$\text{Correlation}(\Delta\text{Formulary}_{diabetes}, \Delta\text{Formulary}_{cardiovascular}) = \rho \tag{12}$$

When ρ exceeds historical norms, hidden nodes activate to signal probable portfolio negotiation, even though no public announcement exists.

5.4.3 Example 3: PBM Margin Pressure Detection

The model learns to recognize when PBMs are under margin pressure and more likely to accept aggressive rebate offers:

- Earnings call language sentiment shifts
- Multiple formulary changes in high-revenue categories
- Unusual timing of negotiation outreach
- Competitor rebate patterns becoming more aggressive

These observables cause specific hidden nodes to activate, signaling opportunity for advantageous negotiation.

5.5 The Information Asymmetry Advantage

Traditional game theory assumes complete or symmetric information. PBM negotiations violate this assumption—information asymmetry is the norm. RBMs thrive in this environment:

$$\text{Strategic Value} = f(\text{Information Asymmetry} \times \text{Hidden Node Learning}) \tag{13}$$

The more opaque the market, the greater the advantage of hidden node inference. While competitors operate blindly, your model provides probabilistic intelligence on:

- **Rebate Benchmarking**: "We are likely 5-8% below competitor rebates in this category"
- **Timing Intelligence**: "PBM is under margin pressure; aggressive offer likely successful"
- **Strategic Detection**: "Competitor preparing major portfolio restructuring based on unusual patterns"
- **Risk Signals**: "High probability of formulary exclusion if we don't improve offer by 3-5%"

5.6 Why Hidden Nodes Succeed Where Other Methods Fail

5.6.1 Consultant Intelligence: Limited and Delayed

Consultants provide quarterly reports based on surveys and relationships. Hidden nodes learn continuously from real-time market data.

5.6.2 Syndicated Data: Backward-Looking

IMS/IQVIA data shows what happened, not why. Hidden nodes infer the "why" from patterns.

5.6.3 Game Theory Models: Require Complete Information

Classical models need inputs you don't have. Hidden nodes learn representations of what you don't know.

5.6.4 Machine Learning Without Hidden Layers: Miss Latent Structure

Standard ML predicts outcomes but doesn't model underlying hidden variables. RBM hidden nodes explicitly represent these latent factors.

The restricted architecture of RBMs assumes these hidden factors explain correlations in observables without requiring direct visible-visible interactions—a reasonable approximation given that market behaviors arise from underlying strategic considerations.

6 Quantum Advantages in Negotiation Space

6.1 Tunneling Through Strategic Barriers

Classical negotiation follows gradient descent in the energy landscape, requiring monotonic improvement:

$$\Delta E_{\text{classical}} < 0 \quad \forall \text{ accepted moves} \tag{14}$$

Quantum tunneling enables transitions through higher-energy intermediate states:

$$P_{\text{tunnel}} \propto \exp\left(-2\int_{x_1}^{x_2} \sqrt{2m(V(x)-E)}dx/\hbar\right) \tag{15}$$

This allows discovery of deal structures requiring temporary concessions before reaching mutual benefit.

6.2 Superposition of Negotiation Strategies

The quantum state exists in superposition across multiple deal structures:

$$|\psi\rangle = \sum_i \alpha_i |s_i\rangle \tag{16}$$

where $|s_i\rangle$ represents distinct negotiation configurations and $\sum_i |\alpha_i|^2 = 1$.

This parallel exploration contrasts with classical sequential evaluation of proposals.

6.3 Nash Equilibria Through Quantum Annealing

The quantum system naturally evolves toward configurations minimizing joint energy, potentially revealing Nash equilibria where:

$$\frac{\partial E_{\text{mfr}}}{\partial s_i} = 0, \quad \frac{\partial E_{\text{pbm}}}{\partial s_j} = 0 \quad \forall i, j \tag{17}$$

7 The Negotiation Process as Simulated Annealing

7.1 Temperature Schedule in Negotiations

The negotiation process naturally follows an annealing schedule:

1. **High Temperature (Initial Phase):** $T = T_{\text{high}}$

$$P(\text{accept}) = \exp(-\Delta E / T_{\text{high}}) \tag{18}$$

   Parties explore radical departures from status quo.

2. **Cooling (Middle Phase):** $T = T_{\text{mid}}$

$$P(\text{accept}) = \exp(-\Delta E / T_{\text{mid}}) \tag{19}$$

   Convergence toward feasible region begins.

3. **Low Temperature (Final Phase):** $T = T_{\text{low}}$

$$P(\text{accept}) = \exp(-\Delta E / T_{\text{low}}) \approx \mathbb{1}[\Delta E < 0] \tag{20}$$

   Only improvements accepted, reaching local minimum.

7.2 Baseline Energy: The BATNA Principle

The reference energy level is established by the Best Alternative to Negotiated Agreement (BATNA):

$$E_{\text{baseline}} = \min\{E_{\text{status quo}}, E_{\text{walk away}}, E_{\text{competitor win}}\} \tag{21}$$

Any acceptable deal must satisfy:

$$E_{\text{deal}} < E_{\text{baseline}} \tag{22}$$

8 Information Asymmetry and Hidden Unit Learning

8.1 The Information Structure

Each party possesses private information creating asymmetries:

- Manufacturers know: $I_{\text{mfr}} = \{\text{R\&D costs, pipeline, production capacity}\}$
- PBMs know: $I_{\text{pbm}} = \{\text{all rebates, plan preferences, utilization}\}$
- Plans know: $I_{\text{plan}} = \{\text{patient outcomes, adherence, total cost}\}$

8.2 Learning Hidden States

The Boltzmann machine learns these hidden states through observable behaviors:

$$P(\mathbf{h}|\mathbf{v}) = \prod_i \sigma(W_i \cdot \mathbf{v} + b_i) \tag{23}$$

where σ is the sigmoid function. The learned hidden representations capture strategic information without explicit access to private data.

9 Complex Interactions in the PBM Ecosystem

9.1 Portfolio-Level Negotiations

The energy function must capture cross-product effects:

$$E_{\text{portfolio}} = \sum_{i,j} \theta_{ij} s_i s_j \tag{24}$$

where θ_{ij} represents interaction strength between products i and j in the portfolio.

9.2 Competitive Dynamics

The presence of competitors modifies the energy landscape:

$$E_{\text{competitive}}(\mathbf{s}) = E_{\text{base}}(\mathbf{s}) + \sum_c \phi_c(\mathbf{s}, \mathbf{s}_c) \tag{25}$$

where $\mathbf{s}_c$ represents competitor positions and ϕ_c captures competitive interactions.

9.3 Regulatory Constraints

Regulatory pressure adds energy penalties for certain configurations:

$$E_{\text{regulatory}}(\mathbf{s}) = \begin{cases} 0 & \text{if compliant} \\ \infty & \text{if violates hard constraints} \\ \rho \cdot \text{risk}(\mathbf{s}) & \text{if gray area} \end{cases} \tag{26}$$

10 Market Phase Transitions

10.1 Metastable States

The current high list price/high rebate equilibrium may represent a metastable state:

$$\frac{\partial^2 E}{\partial s^2} > 0 \quad \text{(local minimum)} \tag{27}$$

but with lower energy states accessible through collective action.

10.2 Critical Points

The system may undergo phase transitions at critical points where:

$$\det\left(\frac{\partial^2 E}{\partial s_i \partial s_j}\right) = 0 \tag{28}$$

These points indicate where small perturbations can trigger market reorganization.

11 Quantum Training of RBMs for PBM Negotiations

11.1 D-Wave's Quantum AI Toolkit

D-Wave's Ocean software suite now provides direct integration between quantum annealers and PyTorch, enabling quantum-assisted training of RBMs.

This toolkit fundamentally changes the implementation landscape by allowing quantum computers to accelerate both the training and sampling phases of RBM operations.

The quantum training process leverages the quantum annealer's ability to:

1. Sample from the Boltzmann distribution at hardware speed
2. Find low-energy configurations through quantum tunneling
3. Escape local minima during gradient estimation

11.2 Quantum Contrastive Divergence

Traditional RBM training uses Contrastive Divergence (CD) to approximate gradients:

$$\Delta w_{ij} = \epsilon(\langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}) \tag{29}$$

Quantum annealing enhances this by providing better samples from the model distribution. The quantum sampler naturally provides samples from:

$$P(\mathbf{v}, \mathbf{h}) \propto \exp(-E(\mathbf{v}, \mathbf{h})/T_{\text{effective}}) \tag{30}$$

where $T_{\text{effective}}$ includes both thermal and quantum fluctuations.

11.3 PyTorch Integration for PBM Data

The integration with PyTorch enables seamless handling of PBM market data:

```
# Pseudocode structure using D-Wave Ocean + PyTorch
import torch
from dwave.plugins.pytorch import QuantumRBM

class PBMNegotiationRBM(QuantumRBM):
    def __init__(self, n_visible, n_hidden, sampler):
        super().__init__(n_visible, n_hidden)
        self.sampler = sampler  # D-Wave quantum sampler

    def quantum_sample(self, v_data):
        # Map to QUBO for quantum annealer
        Q = self.to_qubo(v_data)
```

```
        # Sample using quantum annealing
        response = self.sampler.sample_qubo(Q)
        return self.from_qubo(response)
```

11.4 Advantages of Quantum Training for PBM Applications

11.4.1 Escaping Local Minima in Weight Space

During training on historical negotiation data, classical CD often gets trapped in local minima, learning suboptimal representations. Quantum tunneling during training allows the model to:

$$P_{\text{tunnel}}(w \to w') \propto \exp\left(-\frac{S_{\text{barrier}}}{\hbar}\right) \tag{31}$$

This enables discovery of weight configurations that better capture the true market dynamics.

11.4.2 Sampling from Complex Distributions

PBM negotiations involve multimodal distributions (different types of deals). Quantum sampling efficiently explores these modes:

$$\langle O \rangle_{\text{quantum}} = \sum_{\text{modes}} p_i \langle O \rangle_i \tag{32}$$

where the quantum system naturally weights different modes by their Boltzmann factors.

11.4.3 Learning Hidden Market Structures

The quantum RBM can learn latent structures that classical training might miss:

- Rare but stable deal configurations (deep, narrow minima)
- Transitional states between market regimes
- Correlated patterns across seemingly independent negotiations

12 Implementation Considerations

12.1 Mapping to Quantum Hardware

The negotiation problem must be formulated as a Quadratic Unconstrained Binary Optimization (QUBO):

$$\text{QUBO} : \min_{\mathbf{x} \in \{0,1\}^n} \mathbf{x}^T Q \mathbf{x} \tag{33}$$

This maps to the Ising model implemented in quantum annealers:

$$H_{\text{Ising}} = -\sum_{i,j} J_{ij} s_i s_j - \sum_i h_i s_i \tag{34}$$

where $s_i \in \{-1, +1\}$.

12.2 Hardware Topology Constraints

Current quantum annealers have limited connectivity (Chimera or Pegasus graphs), requiring embedding overhead for fully connected problems. The embedding maps logical qubits to chains of physical qubits:

$$\text{Logical qubit } q_L \rightarrow \text{Chain } \{q_{p1}, q_{p2}, ..., q_{pk}\} \tag{35}$$

12.3 Hybrid Classical-Quantum Training Pipeline

The D-Wave toolkit enables a hybrid approach where:

1. Data preprocessing and feature engineering occur classically
2. Quantum annealing performs sampling and energy minimization
3. PyTorch handles gradient computation and weight updates
4. The process iterates until convergence

This hybrid architecture leverages the strengths of both paradigms:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}_{\text{quantum}}(\theta_t) \tag{36}$$

where $\mathcal{L}_{\text{quantum}}$ uses quantum-sampled statistics.

13 Building Competitive Moats Through Hidden Intelligence

13.1 Why This Creates Sustainable Advantage

Even when competitors eventually adopt similar quantum RBM approaches, first-movers maintain multiple advantages:

13.1.1 Data Network Effects

Your hidden nodes are trained on your unique history:

- Your specific win/loss patterns across negotiations
- Your portfolio's particular competitive dynamics
- Your historical responses to market changes

This creates a model that understands your specific competitive position better than any generic approach.

13.1.2 Learning Curve Advantages

Early adopters accumulate:

- Better intuition for interpreting hidden node patterns
- Refined data collection processes that capture subtle signals
- Validated correlation between hidden node activations and actual outcomes

13.1.3 Strategic Deception Capability

Once you understand how hidden nodes learn from market observables, you can:

- Create false signals to mislead competitor models
- Mask your true strategic intentions
- Exploit competitors who rely on similar models without understanding their limitations

13.2 The Arms Race Dynamic

The adoption of quantum RBMs in PBM negotiations will likely follow an arms race pattern:

$$\text{Advantage}(t) = \begin{cases} \text{High} & \text{Early adoption phase} \\ \text{Moderate} & \text{Competitive parity phase} \\ \text{Negative} & \text{If you lack capability when others have it} \end{cases} \tag{37}$$

This creates urgency: the advantage is greatest for early adopters and becomes a strategic necessity as adoption spreads.

13.3 Quantifying the Intelligence Value

For a pharmaceutical company with $20B in US revenue:

- Average rebate: 40% of list price
- Total rebate spend: $8B annually
- 1% improvement through better intelligence: $80M annual value
- 2-3% improvement (realistic with information advantage): $160-240M

Beyond direct rebate optimization:

- Avoiding formulary exclusions: Each prevented exclusion worth $50-500M
- Earlier competitive response: 3-month earlier detection worth millions in protected share
- Portfolio optimization: Better cross-product leverage worth 2-5% in total rebate efficiency

14 Future Exploration: Quantum AI Toolkit for PBM Negotiations

Over the next several weeks, I plan to conduct a deep, hands-on exploration of D-Wave's Quantum AI Toolkit, with a particular emphasis on its potential for accelerating Restricted Boltzmann Machine (RBM) training in the PBM negotiation context. I'll be running these experiments on a custom-built

system I built with over 3,000 CUDA cores, 112 Tensor Cores, and a Ryzen 9 processor running Ubuntu 22.04 — providing the computational backbone to push this work from concept into hardware-level validation. The goals of this exploration will be to:

- Benchmark quantum-assisted RBM training against classical approaches on representative PBM market datasets.
- Evaluate sampling quality from the quantum annealer in scenarios with complex, multimodal market dynamics.
- Prototype a hybrid training pipeline that integrates quantum sampling with classical gradient updates.
- Identify the most relevant RBM configurations for detecting latent competitive signals in rebate and formulary negotiations.

The outcome of this work will be a set of practical, tested workflows that could form the foundation for quantum-enhanced hidden-intelligence models in PBM rebate strategy. This will also allow me to quantify the gap between current classical model performance and what is achievable with quantum resources.

15 Theoretical Market Implications

15.1 Discovery of Hidden Market Equilibria

Quantum annealing may reveal stable market configurations invisible to classical analysis, where apparent prisoner's dilemmas have cooperative solutions accessible through quantum tunneling.

15.2 Information Revelation Through Energy Minimization

The process of finding low-energy states implicitly reveals information about private valuations, as parties must truthfully signal preferences to reach optimal configurations.

15.3 System-Wide Efficiency

By simultaneously optimizing across multiple stakeholders, quantum Boltzmann machines could identify Pareto improvements that reduce deadweight loss in the pharmaceutical supply chain.

16 Conclusion

This framework represents a fundamental shift in how pharmaceutical manufacturers can approach PBM negotiations. The critical insight is that Restricted Boltzmann Machines, particularly when trained using quantum annealing, transform information asymmetry from a strategic liability into a quantifiable advantage. The hidden nodes in these models naturally learn to represent the very information competitors guard most closely—rebate percentages, portfolio structures, strategic intentions—by inferring their effects from observable market patterns.

The availability of D-Wave's quantum AI toolkit with PyTorch integration makes this more than theoretical speculation. Organizations may now be able to implement quantum-trained RBMs that discover competitive intelligence patterns classical methods would miss. The quantum advantage would manufest in two critical ways: during training, where quantum tunneling enables discovery of rare but valuable patterns hiding in narrow probability wells, and during inference, where quantum sampling explores negotiation strategies inaccessible through classical paths.

For pharmaceutical executives, the implications are clear and immediate:

- **Competitive Intelligence**: Every negotiation can now be informed by probabilistic estimates of competitor rebates and PBM strategies derived from market observations
- **First-Mover Advantage**: Organizations that develop this capability first will negotiate with systematic information advantages over competitors operating with traditional approaches
- **Defensive Necessity**: As competitors develop similar capabilities, lacking this technology becomes a strategic vulnerability
- **ROI Justification**: Even marginal improvements in rebate efficiency through better intelligence can yield hundreds of millions in value for large portfolios

The hidden node architecture directly addresses the core challenge of PBM negotiations: making strategic decisions despite incomplete information. While you will never know your competitor's exact rebate, you can now estimate probability distributions over likely rebate ranges. While you cannot access PBM's internal margin calculations, you can infer their satisfaction from market behaviors. While portfolio deals remain confidential, you can detect their presence through cross-market correlations.

The framework's strength lies not just in its theoretical elegance but in its practical applicability to real negotiation scenarios. Each organization's model becomes unique through training on proprietary historical data, creating sustainable competitive advantage even as the underlying technology becomes widely adopted. The quantum training process ensures that valuable strategic patterns—those rare configurations that could transform negotiation outcomes—are discovered rather than overlooked.

As the pharmaceutical industry faces continued pricing pressure and formulary consolidation, the ability to navigate negotiations with superior intelligence becomes increasingly critical. This framework establishes how quantum computing, applied through RBMs to the challenge of information asymmetry, can provide that intelligence. The question facing pharmaceutical leadership is not whether to pursue this capability, but how quickly it can be deployed before competitors gain the same advantage.

The convergence of quantum computing, machine learning, and deep industry expertise in PBM negotiations creates an unprecedented opportunity. Organizations that recognize and act on this opportunity will find themselves negotiating from positions of information strength, while those that delay will increasingly negotiate blind against better-informed competitors. In a market where every basis point of rebate efficiency matters, the hidden nodes' ability to illuminate what was previously unknowable represents a strategic revolution in pharmaceutical market access.

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Molly Maskrey is a healthcare analytics researcher and quantum computing consultant based in Denver, Colorado. When not developing theoretical frameworks for pharmaceutical optimization, she can be found piloting aircraft over the Rocky Mountains or competing in local tennis tournaments. Currently planning her escape to the San Juan Islands aboard her sailboat, aptly named the *Quantum Nomad*, where she intends to revolutionize healthcare economics one quantum algorithm at a time—with occasional breaks for sailing and shore power-dependent Zoom calls from scenic harbors. Her unique blend of advanced analytics, quantum computing expertise, and questionable life choices makes her the go-to consultant for organizations seeking to bridge the gap between cutting-edge technology and practical business solutions. This work is part of a series exploring theoretical frameworks for real-world quantum applications.